

Reviewer Pack — Minimal Ehrhart Polynomial Degree and Circuit Satisfiability...

Entry 4c3c0f00945a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-06-10 02:32:47 UTC

Minimal Ehrhart Polynomial Degree and Circuit Satisfiability Complexity

Entry ID: 4c3c0f00945a **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-10 02:32:47 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Combinatorics - Ehrhart Theory **Field B** (complexity object): Boolean Circuits - SAT Clause Satisfiability

Statement:

For every CNF φ with n variables, the degree of the minimal non-constant Ehrhart polynomial associated with φ is upper-bounded by a function $f(n)$ that grows as $\Theta(n \log n)$, where $f(n)$ is the maximum number of gates in a circuit that can decide φ .

Rationale (proposer's reasoning):

Ehrhart Theory, a branch of algebraic combinatorics, studies the integer points of polytopes. Its tools could potentially expose nontrivial properties of boolean functions by considering their associated lattice structures. The conjecture suggests that the degree of an Ehrhart polynomial can serve as a measure for circuit complexity.

Taxonomy category: ehrhart_circuit_complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): d10f055a1055f269

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal non-constant Ehrhart polynomial degree for a CNF φ with n variables is upper-bounded by $\Theta(n \log n)$ if all generated lattice points' incidence vectors meet the expected threshold and the aggregate statistic across 30 seeds indicates the degree does not exceed $f(n)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Ehrhart Theory" AND "Boolean Circuits" AND "circuit satisfiability" - "minimal Ehrhart polynomial" AND CNF AND "satisfiability complexity" - "degree of Ehrhart polynomial" AND CNF AND " $\Theta(n \log n)$ "

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(2 * n):
            clause = [random.randint(-n, n) for _ in range(random.randint(1, 3))]
            clauses.append(clause)
        return clauses

    def ehrhart_polynomial_degree(cnf):
        # Placeholder function to simulate Ehrhart polynomial degree calculation
        # This is a dummy implementation and should be replaced with actual computation
        return random.randint(n // 2, n * (n + 1) // 2)

    def circuit_complexity(cnf):
        # Placeholder function to simulate circuit complexity calculation
        # This is a dummy implementation and should be replaced with actual computation
        return random.randint(n // 3, n * (n + 1) // 6)

    n = random.choice([5, 10, 15, 20, 30, 40])
    cnf = generate_cnf(n)
    degree = ehrhart_polynomial_degree(cnf)
    complexity = circuit_complexity(cnf)

    expected_bound = n * (n + 1) // 6

    return {
        "metric_name": "Ehrhart Polynomial Degree",
        "metric_value": degree,
        "instances_tested": 1,
```


The test code crashed before producing data, which means we cannot verify if the conjecture holds or not. | next: Re-run the test with proper error handling to ensure it completes and produces results.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14260
2	preregistration	ollama_remote	glm4:latest	0	0	9890
3	novelty	ollama_remote	glm4:latest	0	0	8573
4	novelty	ollama_remote	glm4:latest	0	0	8987
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	44580
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7065
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8388
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8214
9	judge	ollama_remote	glm4:latest	0	0	25942

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 135898 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/4c3c0f00945a.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/4c3c0f00945a.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/4c3c0f00945a.tar.gz (if generated)